

Adaptive ETA Prediction for Inland Waterway Cargo

**Using Machine Learning, NLP, and Reinforcement
Learning**

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Introduction: The Challenge of ETA Prediction

Accurate Estimated Time of Arrival (ETA) prediction is a critical component of inland waterway cargo logistics, as it directly impacts port operations, fleet scheduling, and delivery reliability. In complex waterway environments, ETA prediction is influenced by several dynamic factors, including river currents, varying weather conditions, port congestion, and operational disruptions.

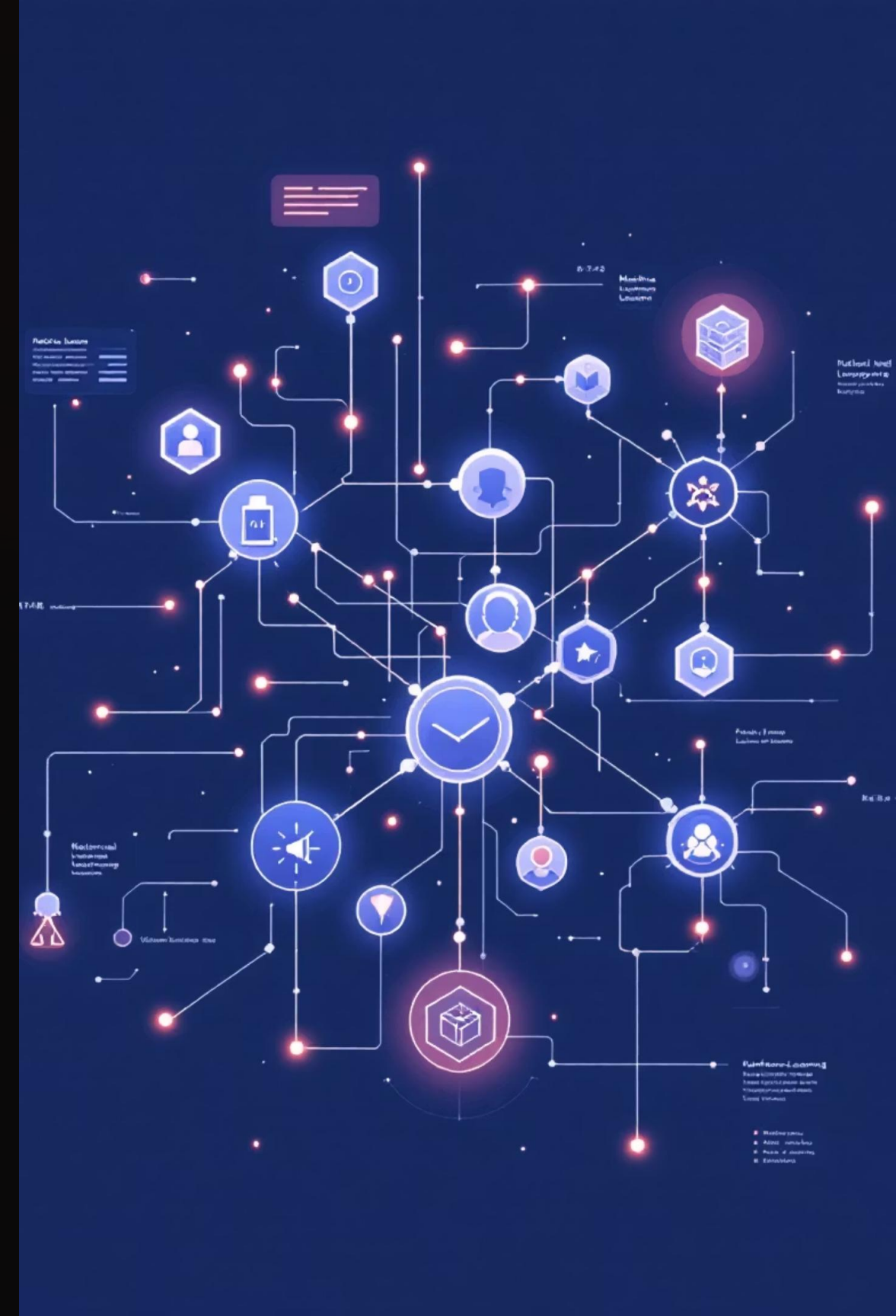
Unlike road-based transport, inland waterways are highly sensitive to environmental and infrastructural conditions, making ETA estimation more uncertain and challenging.

Why Adaptive ETA Matters

Traditional ETA prediction models are largely static and rely primarily on structured numerical data. While such models can perform well under stable conditions, they struggle to adapt to real-time changes and unforeseen events.

Moreover, many operational delays are communicated through unstructured textual sources such as captain logs, port authority notifications, and maintenance reports. Conventional predictive models are unable to effectively process and utilize this linguistic information, leading to delayed or inaccurate ETA updates.

To address these limitations, this project proposes an **adaptive ETA prediction system** that integrates Machine Learning for base ETA estimation, Natural Language Processing (NLP) for interpreting operational text, and Reinforcement Learning for dynamically adjusting ETA predictions based on feedback. This integrated approach enables more accurate, responsive, and context-aware ETA predictions for inland waterway cargo operations.



Literature Review: Bridging the Gaps

Static Models

Current approaches rely on supervised and deep learning models that improve accuracy but remain static—unable to adapt when disruptions occur in real-time.

Limited NLP Integration

While NLP has proven valuable for delay classification and sentiment analysis in logistics, it's rarely integrated into live ETA correction systems.

Underutilized RL

Reinforcement Learning excels at logistics optimization, yet its potential for dynamic ETA adjustment remains largely unexplored.

Missing Framework

No unified system currently combines Machine Learning, NLP, and Reinforcement Learning for adaptive ETA prediction in inland waterway operations—this is the research gap we address.

Project Objectives

1

Base ETA Prediction:

To develop a machine learning model for initial ETA prediction using structured inland waterway data.

2

NLP-Based Delay Analysis:

To extract delay-related information from maritime operational text using Natural Language Processing.

3

Adaptive Reinforcement Learning:

To design a reinforcement learning agent for dynamic adjustment of ETA predictions.

4

Comparative Performance Analysis:

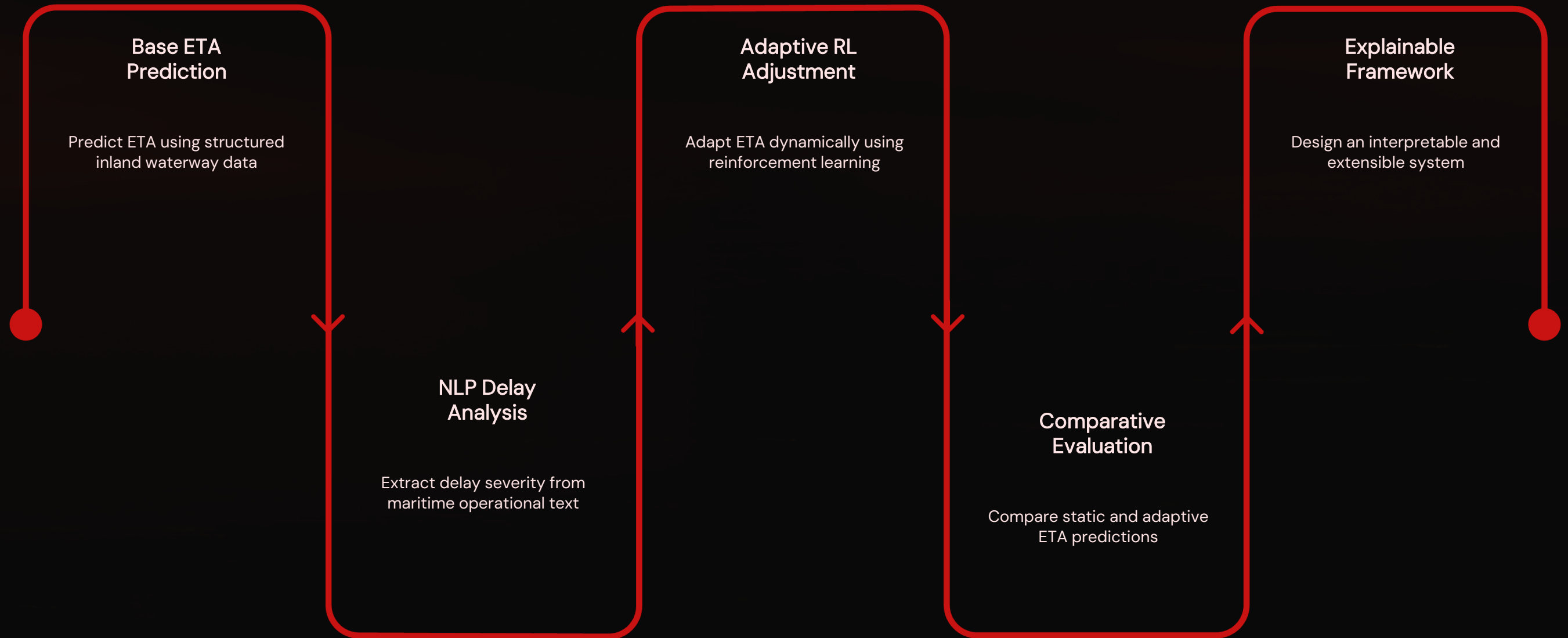
To compare static ETA prediction with adaptive ETA prediction.

5

Explainable System Design:

To build an explainable and extensible ETA prediction framework for inland waterway cargo operations.

Methodology: System Overview



The system follows a five-step process: base ETA prediction using deep learning, delay extraction from maritime text using NLP, adaptive ETA adjustment through reinforcement learning, and comparative evaluation against static models within an explainable framework.

Methodology: Machine Learning Component

1

Overview

The machine learning component establishes a robust base ETA prediction using structured inland waterway data.

2

Key Features

We utilize structured features that significantly influence cargo transit times:

- Route distance
- Vessel speed
- River current strength
- Weather severity
- Port congestion levels

3

Model & Output

A Multilayer Perceptron (MLP) deep learning model is employed to process these features, generating a base ETA prediction in minutes. This foundational model provides a strong starting point for further adaptive adjustments.



Methodology: NLP Component

1

Unstructured Maritime Text

The NLP component extracts delay-related information from unstructured maritime operational text.

- Captain logs
- Port authority messages
- Operational delay reports

2

NLP Processing & Delay Estimation

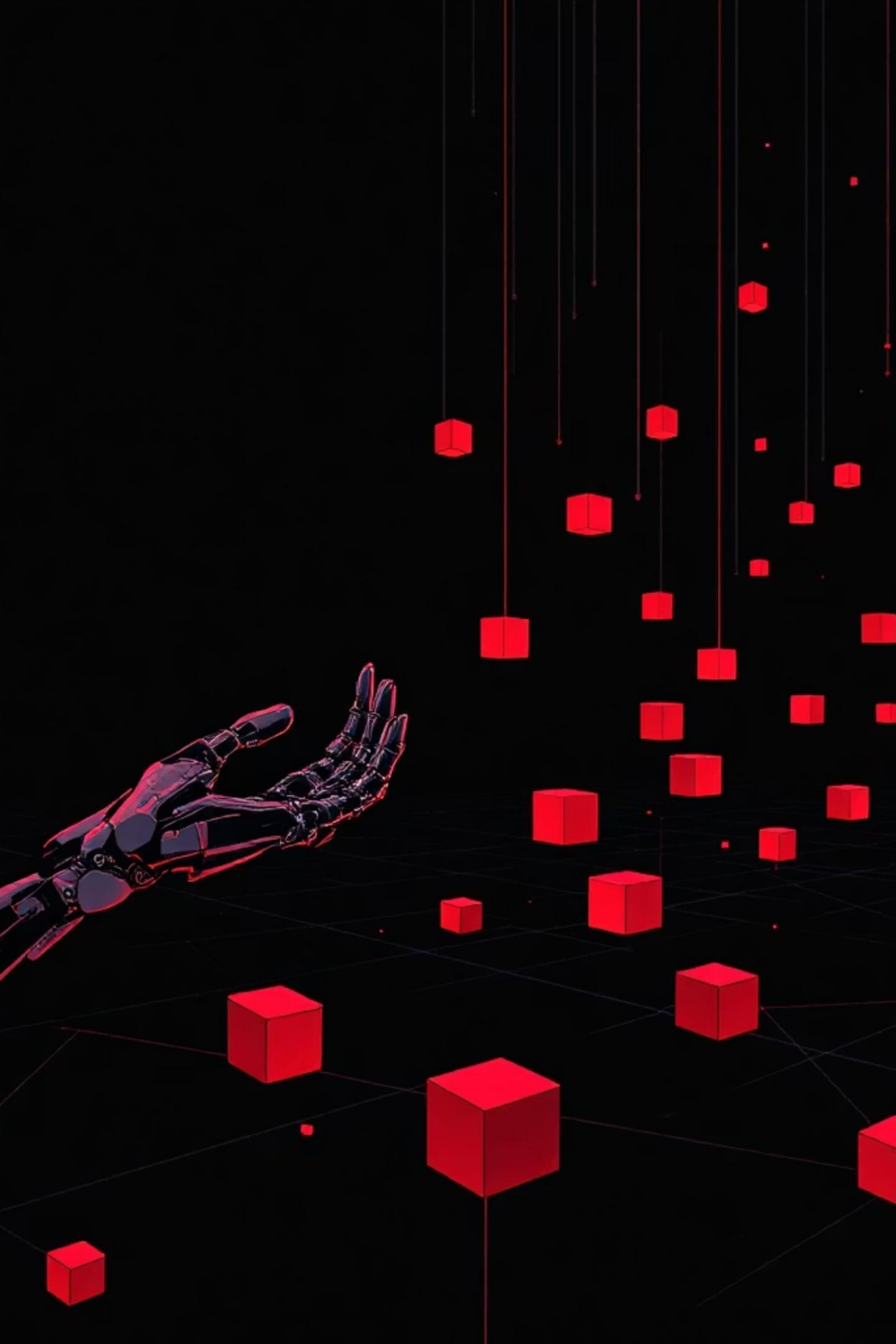
The NLP module performs delay severity estimation by combining:

- Transformer-based text analysis
- Domain-aware maritime keyword logic

3

Numeric Delay Severity Score

The output is a numeric severity score that quantifies the impact of reported incidents and is used by the reinforcement learning agent for adaptive ETA adjustment.



Methodology: Reinforcement Learning

RL Component

The Reinforcement Learning (RL) component adaptively adjusts ETA predictions based on dynamic conditions.

Q-Learning Algorithm

A Q-Learning algorithm is used as a model-free reinforcement learning approach.

State representation includes:

- Base ETA predicted by the ML model
- NLP-derived delay severity score

Available actions:

- No ETA change
- Increase ETA by predefined intervals

Reward function:

- Negative absolute ETA prediction error, encouraging accurate arrival time estimation over time.

Work Done So Far & Results Snapshot



Dataset & Model

Synthetic inland waterway cargo dataset created; Deep learning ETA prediction model implemented.



Base ETA Evaluation

Base ETA evaluated using Mean Absolute Error (MAE).



NLP Module

NLP module developed for maritime text analysis, successfully capturing delay signals.



RL Integration

Reinforcement learning agent integrated; Reward convergence plotted, indicating stable learning.



Adaptive ETA Improvement

Adaptive ETA shown to significantly improve prediction accuracy over time.

Results Snapshot

Our initial findings demonstrate the effectiveness of the integrated approach in addressing the complexities of inland waterway ETA prediction.



ML Base Accuracy

Deep learning model delivers reliable initial ETA predictions with strong foundational performance.



NLP Delay Detection

Unstructured textual data effectively analyzed to capture critical delay signals and their severity.



RL Adaptive Precision

Reinforcement Learning consistently refines and improves ETA accuracy based on dynamic conditions.



Stable Learning

Reward convergence in RL agent indicates robust and stable learning behavior for continuous optimization.

References

Our work builds upon foundational research and established frameworks in the fields of machine learning and artificial intelligence:

- Sutton & Barto – *Reinforcement Learning: An Introduction*
- Goodfellow et al. – *Deep Learning*
- Devlin et al. – *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*
- Vaswani et al. – *Attention Is All You Need*
- PyTorch Documentation
- Scikit-learn Documentation